

Deep Diving into Synthetic Control Method

From Abadie (2010) to Xu (2017)

Sheng-Yuan Chen

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1 The Problem: When Parallel Trends Fails

Difference-in-differences (DiD) is the workhorse of policy evaluation. The setup is familiar: one group gets treated, another does not, and we compare the change in outcomes across the two groups. The identifying assumption is **parallel trends**—in the absence of treatment, the treated group would have followed the same trajectory as the control group.

This assumption is often wrong.

Abadie, Diamond, and Hainmueller (2010) open their paper with exactly this concern. In 1988, California passed Proposition 99—a ballot initiative that raised cigarette taxes by 25 cents per pack and directed the revenue toward anti-smoking campaigns. The question is straightforward: did this reduce cigarette consumption? A naive DiD would compare California’s post-1988 consumption trend against other U.S. states. But California was already an outlier. Its per-capita cigarette sales had been declining faster than the rest of the country since the late 1970s. Using any single state—or even the average of all other states—as the control group would confound the policy effect with whatever structural difference was already driving California’s exceptionalism.

This is the core failure mode of DiD in comparative case studies: the treated unit is often treated *because* it is different. Its pre-treatment trajectory is different. Its demographics are different. Assuming that the control group’s trend would have applied to the treated unit is not an identifying assumption—it is a hope.

More formally, let $Y_{it}(0)$ denote the potential outcome for unit i at time t in the absence of treatment. DiD identifies the ATT under:

$$\mathbb{E}[Y_{it}(0) - Y_{i,t-1}(0)] = \mathbb{E}[Y_{jt}(0) - Y_{j,t-1}(0)] \quad (1)$$

for treated unit i and control unit j . When this fails, the DiD estimate inherits the pre-existing trend gap as bias.

The key question is: can we construct a better counterfactual for California?

The intuition is simple. Instead of using a single control state, we use a *weighted combination* of control states—one that, in the pre-treatment period, closely resembles California on per-capita cigarette sales and on the covariates that predict them. If we can find such a combination, its post-1988 path is a plausible estimate of what California’s cigarette consumption *would have been* had Proposition 99 never passed.

This is the core idea of the **Synthetic Control Method** (Abadie et al. 2010).

The method does not assume parallel trends. Instead, it asks: what weighted average of control units best approximates the treated unit in the pre-treatment period? That weighted average—the “synthetic control”—serves as the counterfactual.

The appeal is partly diagnostic. You can *see* how well the synthetic control tracks the treated unit before the policy. If the fit is poor, the counterfactual is not credible, and the method tells you so explicitly. This transparency is one reason SCM displaced DiD as the default tool for comparative case studies.

The same logic applies whenever a clearly-dated policy affects a small number of units. Taiwan legalized same-sex marriage in May 2019—the first country in Asia to do so. Examining whether this reduced suicide rates requires a counterfactual for Taiwan’s post-2019 trajectory. Taiwan’s pre-treatment suicide trend looks nothing like Japan’s or South Korea’s; constructing a synthetic Taiwan from a weighted combination of regional neighbors is far more credible than any two-country DiD.

But the original SCM has three limitations that matter in practice:

1. It handles only **one treated unit**
2. It produces no standard errors—inference relies on permutation tests
3. It requires a long pre-treatment period to achieve good pre-treatment fit

The Generalized Synthetic Control Method (Xu 2017) addresses all three. Before turning to Xu’s proposal, the next section reviews how the original SCM works mechanically.

2 The Original SCM: Mechanics

The synthetic control method solves a matching problem. We observe $J + 1$ units over T periods. Unit 1 is treated starting at period $T_0 + 1$; units $2, \dots, J + 1$ form the donor pool and remain untreated throughout. The goal is to estimate what unit 1’s outcome *would have been* in the post-treatment periods had the policy never occurred.

The synthetic control for unit 1 is a weighted average of donor units:

$$\hat{Y}_{1t}(0) = \sum_{j=2}^{J+1} w_j^* Y_{jt}, \quad t > T_0 \tag{2}$$

where the weights $w_j^* \geq 0$ sum to one. The estimated treatment effect at each post-treatment period is:

$$\hat{\delta}_{1t} = Y_{1t} - \hat{Y}_{1t}(0) \quad (3)$$

2.1 Choosing the Weights

The method selects $W^* = (w_2^*, \dots, w_{J+1}^*)$ to minimize pre-treatment discrepancy in both the outcome variable and a set of pre-treatment predictors. Formally, the outer optimization minimizes:

$$\|X_1 - X_0 W\|_V = \sqrt{(X_1 - X_0 W)' V (X_1 - X_0 W)} \quad (4)$$

where X_1 is the vector of pre-treatment predictors for the treated unit, X_0 is the corresponding matrix for the donor pool, and V is a diagonal matrix of predictor weights chosen to minimize out-of-sample prediction error.

In the California example, X_1 includes pre-1988 averages of per-capita cigarette sales, income, retail price of cigarettes, and beer consumption. The resulting synthetic California puts most of its weight on Colorado (0.164), Montana (0.199), Nevada (0.234), and Utah (0.334)—states that together closely approximate California’s pre-Proposition-99 trajectory.

The pre-treatment fit is the method’s first diagnostic. If no convex combination of donor units can approximate the treated unit’s pre-treatment path, the synthetic control is not credible. In the California application, the synthetic control closely tracks actual California’s cigarette sales from 1970 to 1988. The post-1988 gap—synthetic California continues on its pre-trend path while actual California’s sales drop sharply—is the estimated effect of Proposition 99. By 2000, California consumed roughly **26 fewer cigarettes per capita per year** than the synthetic counterfactual suggests it would have.

2.2 Three Limitations

The elegance of SCM obscures its constraints.

First, the method is designed for a **single treated unit**. When multiple units adopt a policy at different times—as with U.S. states adopting Election Day Registration laws—a separate synthetic control must be constructed for each, with no unified

framework for combining estimates or testing hypotheses across them.

Second, inference is weak when the donor pool is small. The p-value from the permutation test is bounded below by $1/(J + 1)$. With six donor countries, the minimum achievable p-value is $1/7 \approx 0.143$, regardless of how large the estimated effect is.

Third, the method requires a **long pre-treatment period**. With too few pre-treatment observations, the weights overfit and the post-treatment counterfactual is unreliable.

These are not minor caveats. They bind precisely in the settings where comparative case studies are most common: few units, short panels, multiple treated observations. Xu (2017) proposes a method that addresses all three—by recognizing that SCM is a special case of a much more general model.

3 What GSC Adds: The Interactive Fixed Effects Model

The original SCM works well when one unit is treated and the donor pool is rich. But the method has no explicit model underlying it—it is a reweighting scheme, not a structural model. Xu (2017) asks: what statistical model does SCM implicitly assume? The answer motivates a generalization.

3.1 From TWFE to IFE

The standard two-way fixed effects (TWFE) model writes:

$$Y_{it} = \delta D_{it} + \alpha_i + \xi_t + \varepsilon_{it} \tag{5}$$

where α_i is a unit fixed effect and ξ_t is a time fixed effect. The parallel trends assumption is baked into the additive structure: every unit’s untreated potential outcome shifts by the same ξ_t in each period. This is why TWFE fails when units respond differently to common shocks—for instance, when a global recession hits California harder than Montana because of differences in industry composition.

The **Interactive Fixed Effects (IFE)** model replaces the additive time effect with an interactive term:

$$Y_{it} = \delta_{it}D_{it} + x'_{it}\beta + \lambda'_i f_t + \varepsilon_{it} \quad (6)$$

The key objects are:

- $f_t \in \mathbb{R}^r$: a vector of **common factors** that evolve over time. These capture aggregate shocks—business cycles, global trends, institutional changes—that affect all units but not necessarily equally.
- $\lambda_i \in \mathbb{R}^r$: **factor loadings** specific to each unit. They measure how sensitive unit i is to each common factor.
- $\lambda'_i f_t$: the interaction. A unit with large loadings on a factor will be strongly affected when that factor moves.

This is a strictly more general model than TWFE. TWFE is the special case where $r = 1$ and all λ_i are equal—meaning every unit reacts identically to the common time trend. The IFE model relaxes this: units can have heterogeneous reactions to common shocks, which is exactly what drives parallel trends violations in practice.

3.2 Why SCM Is a Special Case

Xu (2017) shows that SCM implicitly estimates a version of the IFE model by constructing a synthetic control whose factor loadings match those of the treated unit. When the pre-treatment fit is perfect, the synthetic control and the treated unit share the same λ_i , so their post-treatment paths would be identical absent treatment. The IFE model makes this logic explicit and extendable.

The generalization matters because the IFE framework accommodates what SCM cannot: multiple treated units, treatment at different times, and standard errors derived from a coherent probability model rather than a permutation argument.

4 Estimation: The Three-Step Algorithm

The GSC estimator proceeds in three steps, using only pre-treatment data on control units to learn the factor structure, then projecting it onto treated units to construct counterfactuals.

Step 1: Estimate the factor model on control units. Using only the untreated observations (all control units, all periods; treated units in pre-treatment periods), estimate the IFE model by minimizing:

$$\sum_{(i,t):D_{it}=0} (Y_{it} - x'_{it}\beta - \lambda'_i f_t)^2 \quad (7)$$

This yields estimates $\hat{\beta}$, $\hat{f}_t$, and $\hat{\lambda}_i$ for all control units. The number of factors r is selected by cross-validation (described below).

Step 2: Recover factor loadings for treated units. For each treated unit i , use its pre-treatment outcomes to find $\hat{\lambda}_i$ that best fits the estimated factor path:

$$\hat{\lambda}_i = \arg \min_{\lambda} \sum_{t \leq T_0} \left(Y_{it} - x'_{it}\hat{\beta} - \lambda' \hat{f}_t \right)^2 \quad (8)$$

This is a simple OLS projection of the treated unit’s pre-treatment residuals onto the estimated factors.

Step 3: Impute counterfactuals and estimate ATT. The counterfactual outcome for treated unit i in post-treatment period t is:

$$\hat{Y}_{it}(0) = x'_{it}\hat{\beta} + \hat{\lambda}'_i \hat{f}_t \quad (9)$$

The treatment effect for unit i at period t is $\hat{\delta}_{it} = Y_{it} - \hat{Y}_{it}(0)$, and the ATT averages over all treated units and post-treatment periods. Because the method estimates a separate $\hat{\lambda}_i$ for each treated unit, it naturally delivers **unit-level heterogeneous treatment effects**—something the original SCM cannot do when multiple units are treated.

4.1 Selecting the Number of Factors

How many factors r should the model include? Too few and the factor structure underfits, leaving systematic variation in the residuals. Too many and the model overfits the pre-treatment data, producing unreliable counterfactuals.

Xu (2017) selects r via cross-validation on control units. The procedure holds out one control unit at a time, estimates the IFE model on the remaining controls, and measures how well it predicts the held-out unit’s outcomes. The r that minimizes average prediction error across all held-out units is selected. This is a data-driven

choice with no researcher discretion.

4.2 Inference via Parametric Bootstrap

Unlike SCM’s permutation test, GSC produces genuine standard errors. Xu (2017) uses a parametric bootstrap: draw new errors ε_{it}^* from the estimated residual distribution, construct bootstrap outcomes $Y_{it}^* = \hat{Y}_{it}(0) + \hat{\delta}_{it}D_{it} + \varepsilon_{it}^*$, re-estimate the model, and re-compute the ATT. Repeating this 1,000 times yields a bootstrap distribution from which confidence intervals are read off directly. The bootstrap is blocked at the unit level to preserve within-unit serial correlation.

5 When Does GSC Beat TWFE? Monte Carlo Evidence

Xu (2017) validates the method with a set of Monte Carlo experiments designed to mimic realistic TSCS data. The data-generating process follows the IFE model with two latent factors, heterogeneous factor loadings, and either constant or heterogeneous treatment effects. The benchmark is a two-way fixed effects estimator and the standard IFE estimator (Bai 2009), which imposes a constant treatment effect.

The central finding is easy to state: **when time-varying confounders are present, TWFE is badly biased and GSC is not.**

With decomposable time-varying confounders (i.e., the untreated potential outcome follows an IFE structure), TWFE bias is large and grows with the strength of the factor loadings. The IFE estimator with a known number of factors is unbiased, but in practice we do not know r . The GSC estimator, using cross-validation to select r , is nearly unbiased across all configurations Xu considers.

Two caveats are important. First, when T_0 is small (fewer than 10 pre-treatment periods) or N_{co} is small (fewer than 40 control units), all estimators perform worse, and GSC’s advantage shrinks as the factor structure becomes harder to identify. Second, when parallel trends holds exactly—so the true DGP is additive—TWFE and GSC produce similar estimates, but GSC pays a small efficiency cost from estimating unnecessary factors. This is an acceptable price: you rarely know in advance whether parallel trends holds, and GSC fails gracefully when it does.

6 Empirical Illustration: EDR Laws and Voter Turnout

Xu (2017) applies GSC to estimate the effect of Election Day Registration (EDR) laws on voter turnout in U.S. states. EDR allows citizens to register on Election Day itself, removing a procedural barrier to voting. Nine states adopted EDR at different points between 1973 and 2008—a staggered adoption design that SCM cannot handle in a unified way.

The sample covers all 50 U.S. states from 1920 to 2014, with 38 states serving as the donor pool. Cross-validation selects $r = 2$ factors. The first factor captures the sharp increase in southern states’ turnout following the 1965 Voting Rights Act—a major structural break concentrated in a subset of states. The second reflects a broad downward trend in turnout across recent decades. Both are meaningful and interpretable, which is a useful sanity check on the method.

The overall ATT estimate is approximately +5 **percentage points** (standard error 2.3 pp), consistent with existing large-sample evidence from individual-level data. The diagnostic that matters most here is the factor loadings plot: the nine treated states’ estimated loadings lie mostly within the convex hull of the control states’ loadings, which means the counterfactuals are produced by interpolation rather than extrapolation. This is the GSC analog of checking pre-treatment fit in SCM.

6.1 Heterogeneous Effects Across Adoption Waves

The most interesting result is not the average effect but the heterogeneity. States adopted EDR in three waves, and their motivations differed:

Wave	States	ATT (pp)	Significant?
1st (mid-1970s)	ME, MN, WI	+7.3	Yes
2nd (NVRA-avoiders)	ID, NH, WY	+2.2	No
3rd (recent)	MT, IA, CT	−1.1	No

Early adopters (Maine, Minnesota, Wisconsin) enacted EDR because officials genuinely wanted higher turnout—and turnout rose by about 7 percentage points. Later adopters introduced EDR primarily to opt out of the 1993 National Voter Registration Act (NVRA), which they viewed as administratively costly. In these states, EDR had no detectable effect. The NVRA had already lowered registration barriers substantially, leaving little room for EDR to add further.

This pattern—a large effect for early adopters, near-zero for later ones—is consistent with selection on motivation. It is also exactly the kind of result that a pooled TWFE estimate would obscure, since it averages over states with very different treatment dynamics.

7 Practical Checklist: When to Use GSC

GSC is not always the right tool. Here is a honest accounting of when it works, when it struggles, and what to check before trusting the results.

7.1 Prerequisites

Condition	Recommendation
$T_0 \geq 10$ pre-treatment periods	Safe to proceed
$N_{co} \geq 40$ control units	Safe to proceed
$T_0 < 10$ or $N_{co} < 40$	Results may be unreliable; use with caution
Parallel trends holds approximately	DiD/TWFE is fine and more efficient
Multiple treated units	GSC preferred over running separate SCMs
One treated unit	Original SCM or GSC both work
Staggered adoption timing	GSC handles it natively
Dynamic treatment effects	GSC accommodates them; TWFE does not

7.2 Three Diagnostics to Always Run

1. Pre-treatment fit. Plot actual and imputed outcomes for treated units in the pre-treatment period. If the GSC counterfactual tracks the treated unit well before the policy, the post-treatment gap is credible. If not, stop.

2. Factor loadings plot. Plot the estimated $\hat{\lambda}_i$ for all units in the two-dimensional factor space. Check that the treated units' loadings fall within or near the convex hull of the control units' loadings. If treated units are far outside the hull, the counterfactuals rely on extrapolation and the estimates are unreliable.

3. Sensitivity to r . Re-run the analysis with $r - 1$ and $r + 1$ factors. If the estimated ATT changes substantially, the result is not robust to the factor selection choice.

7.3 Known Limitations

Xu (2017) is explicit about what GSC cannot handle. The method assumes a linear IFE structure for the untreated potential outcome. It does not accommodate dynamic relationships between treatment and outcomes, structural breaks in the factor process, or random coefficients on observed covariates. It also cannot allow treatment timing to be endogenous to the outcome in a continuous way— the policy must be a discrete, clearly-dated event.

When these conditions are violated, the results may be misleading, and the diagnostics above may not flag the problem. In such cases, the honest choice is to fall back on a simpler, more transparent estimator and acknowledge the identification limitations directly.

The bottom line. GSC fills a genuine gap: it gives researchers a principled way to construct counterfactuals in settings where DiD’s parallel trends assumption is implausible and SCM’s single-unit restriction is binding. The method is not magic—it replaces one set of assumptions with another. But the assumptions are more credible in many applied settings, and the diagnostics make it clear when they are not.